



MOBILE APP ECOSYSTEM

Hop & Barley

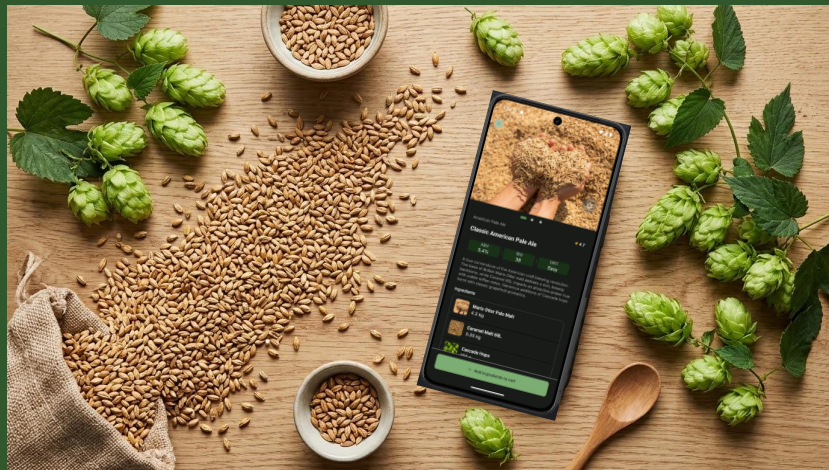
The Complete Ecosystem for Craft Homebrewers

An all-in-one mobile ecosystem connecting craft beer recipes directly with fresh brewing ingredient shopping.

PRESENTER

Andrii Toptun

React Native / Mobile Developer



Bridging the Gap Between Recipe Discovery & Ingredient Procurement

THE PROBLEM

Fragmented Discovery & Calculations

The homebrewing market suffers from a strict disconnect between finding recipes on forums or calculators and purchasing ingredients across separate web shops.

Beginners struggle to configure precise brewing parameters—such as **attenuation**, **IBU units**, and **yeast pitching rates**—without unified guidance.

THE SOLUTION

Unified Mobile Exploration & Checkout

Hop & Barley delivers a cohesive mobile platform that seamlessly bridges recipe exploration with instant ingredient checkout.

By automating complex calculations and streamlining procurement into a single tap, it simplifies the entire brewing process for users of all skill levels.

Delivering a Comprehensive E-Commerce and Recipe Platform

SECURITY & ACCESS

Auth & Security

Full registration flow featuring 4-digit OTP verification with a timer, password reset, and guest browsing capabilities.

INVENTORY & SEARCH

Store Catalog

Live search with multi-category filters for Yeast, Hops, Malts, and Adjuncts, alongside 5 distinct sorting modes.

BREWING GUIDES

Recipe Library

Categorized by Easy, Medium, and Hard difficulty levels with ABV/IBU badges and step-by-step mashing, boiling, and fermentation guides.

PURCHASING FLOW

Cart & Checkout

Dynamic item counter, instant toast feedback, and comprehensive checkout order summaries.

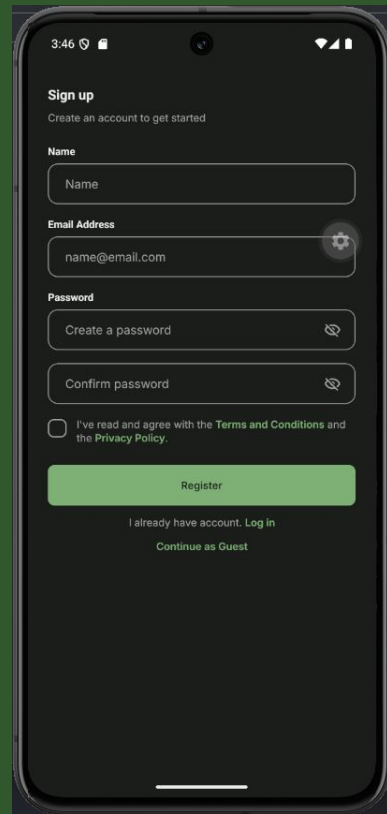
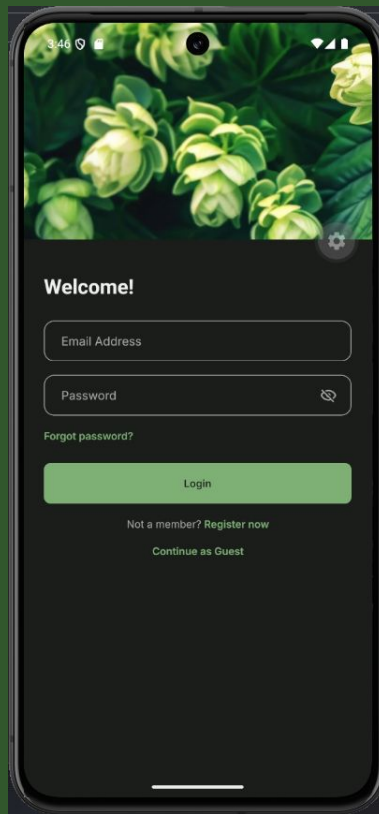
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Auth & Security

Full registration flow featuring 4-digit OTP verification with a timer, password reset, and guest browsing capabilities.

- **OTP Verification:** 4-digit code with live countdown timer
- **Password Recovery:** Automated reset via secure email flow
- **Guest Mode:** Browse store catalog without forced signup
- **Session State:** Encrypted token storage via AsyncStorage



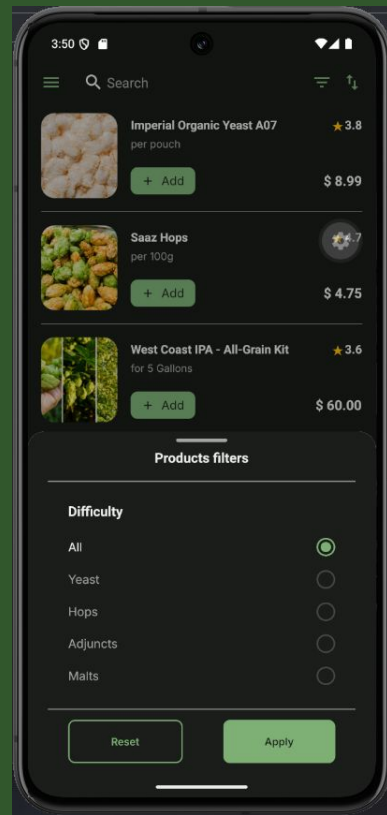
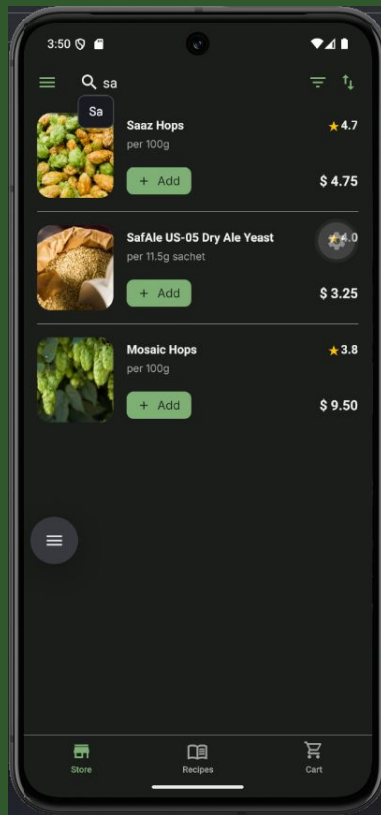
INVENTORY & SEARCH

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Store Catalog

Live search with multi-category filters for Yeast, Hops, Malts, and Adjuncts, alongside 5 distinct sorting modes.

- **Multi-Category Filters:** Quick filtering for Yeast, Hops, Malts, and Adjuncts
- **5 Sorting Modes:** Sort by Name (A-Z / Z-A), Price, or Rating
- **Live Search:** Instant query matching across entire store catalog
- **Quick Add:** One-tap add to cart directly from item list



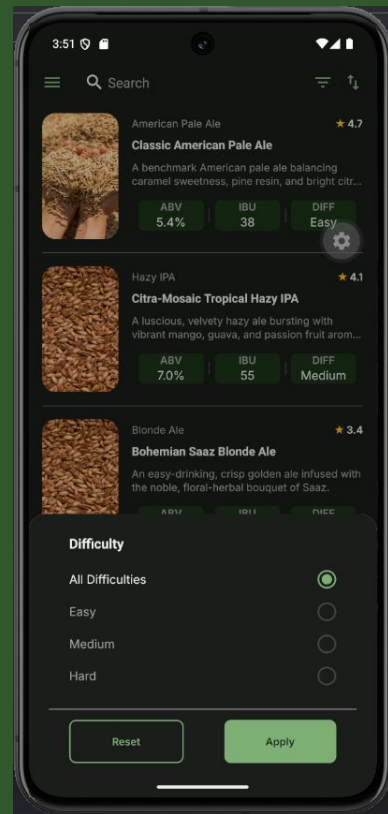
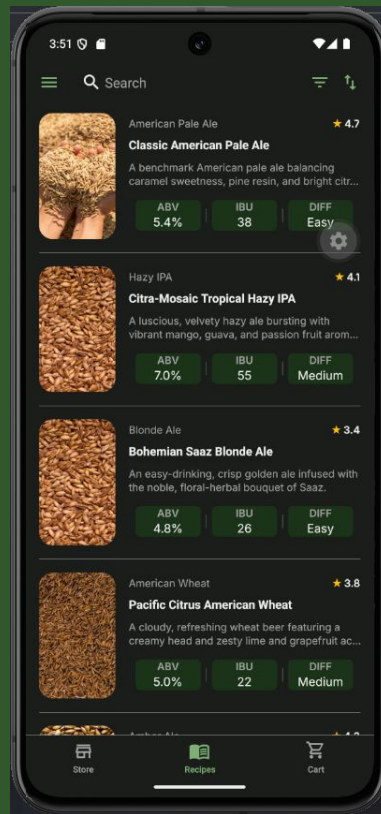
BREWING GUIDES

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Recipe Library

Categorized by Easy, Medium, and Hard difficulty levels with ABV/IBU badges and step-by-step mashing, boiling, and fermentation guides.

- **Difficulty Levels:** Categorized into Easy, Medium, and Hard
- **Vital Stats Badges:** Instant visibility for ABV, IBU, and style tags
- **Step-by-Step Guides:** Detailed mashing, boiling, and fermentation instructions
- **Sorting & Search:** Quick filtering by rating, A-Z, or complexity



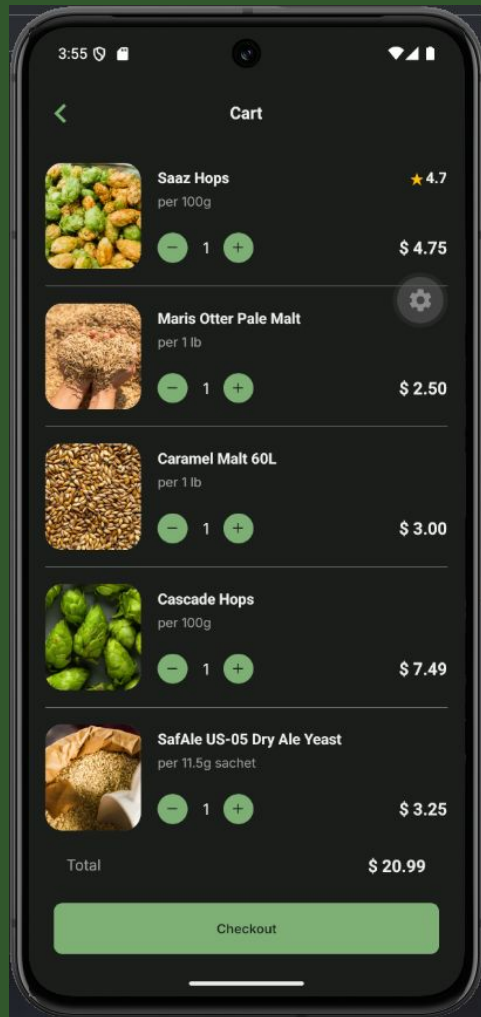
PURCHASING FLOW

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Cart & Checkout

Dynamic item counter, instant toast feedback, and comprehensive checkout order summaries.

- **Dynamic Item Counter:** Real-time price recalculation and quantity controls
- **Instant Toast Feedback:** Immediate status updates on cart changes
- **Order Summaries:** Detailed itemized list with subtotal and total breakdown
- **Seamless Checkout:** One-tap transition to final order placement



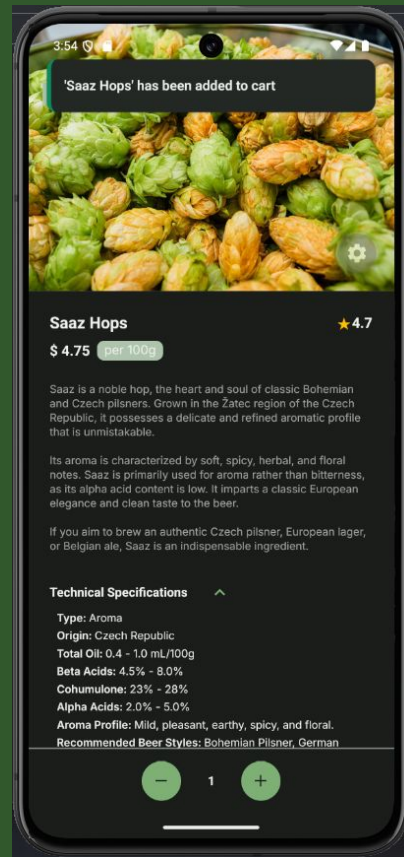
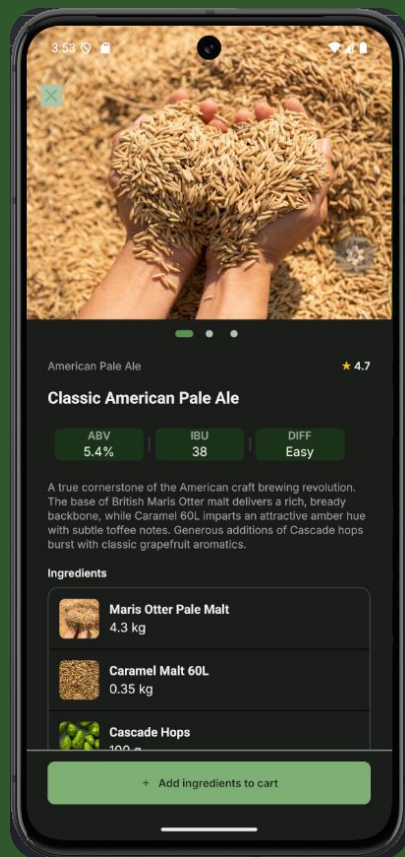
READY-TO-BREW KITS

1-CLICK CONVERSION

Recipe Kits & Technical Specs

Seamlessly converting custom and catalog recipes directly into ready-to-brew ingredient kits with deep technical specs.

- **1-Click Recipe Kits:** Pushes all ingredients (malts, hops, yeast) directly into the cart with a single tap.
- **Deep Technical Specs:** Detailed sheets detailing flocculation, attenuation, temp ranges, and alpha acids.
- **Automated Quantities:** Exact weight scaling per batch size requirements.
- **Instant Checkout Sync:** Directly connected to inventory and cart workflows.



Built with Modern React Native and Expo Technologies

React Native

Core Framework

React Native framework robustly powered by Expo for unified cross-platform development and rapid deployments.

State & Persistence

Centralized store combined with AsyncStorage for dependable local data persistence, session management, and theme preferences.

Expo

React Navigation

Navigation Architecture

React Navigation utilizing Bottom Tabs for main screens, a Drawer for settings & profiles, and a Stack for modals and auth flows.

UI / UX Experience

Supports Dark, Light, and System themes with instant switching, responsive adaptive layouts, and integrated toast alerts for user feedback.

AsyncStorage

Overcoming Complex State & Performance Hurdles



State Management

Cart State Synchronization

Ensuring atomic updates across the catalog, detailed product views, and the cart list to maintain accurate inventory and pricing.



Performance

Filter & Sort Pipeline

Implementing low-latency, client-side query filtering that prevents unnecessary component re-renders during catalog browsing.



UX & Auth

OTP Flow

Designing an accessible pin-code input system with smooth auto-focus transitions and robust resend timer logic.

Expanding the Hop & Barley Ecosystem



Phase 01

In-App Brewing Calculator

Upcoming features for OG/FG, ABV, and IBU calculations directly within the application.



Phase 02

Community & Custom Recipes

Enabling user-created recipe saving and sharing to foster a community of brewers.



Phase 03

Payment Gateway

Planned integration with Stripe, Apple Pay, and Google Pay for seamless transactions.

Thank you for your time.